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FINAL PROJECT REPORT

PREDICTING HDB RESALE FLAT PRICES TO SUPPORT PROPERTY PRICING DECISIONS IN SINGAPORE





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ABSTRACT

This project develops a data-driven decision-support system for Singapore Housing and Development Board (HDB) resale price analysis and prediction. The study uses 52,916 resale transactions from 2024 to 2025, while data from January to June 2026 is reserved for independent testing.

MySQL and KNIME were used for data quality assessment, preparation, and predictive modelling. The original dataset was retained, with 51,635 transactions classified as model candidates and 1,281 transactions flagged for further review. A Linear Regression model was trained using 2024 data and evaluated on 24,286 validation transactions from 2025.

The model achieved an R^2 of 0.830, a Mean Absolute Error (MAE) of SGD 54,923.05, and a Root Mean Squared Error (RMSE) of SGD 72,639.19 on the validation dataset. It also performed better than the selected Median Baseline. On the independent 2026 test dataset of 12,226 transactions, the model recorded an MAE of SGD 59,316.49. An empirical prediction range achieved 88.34% coverage.

The results were presented through interactive Tableau dashboards to support three practical recommendations: using model predictions alongside historical transactions, communicating pricing uncertainty through estimated ranges, and prioritising manual review of higher-risk property segments. The proposed system supports more informed pricing decisions while recognizing the limitations of predictive models.

1. INTRODUCTION AND PROBLEM DEFINITION

1.1. Background and Problem Context

Public housing developed by Singapore's Housing & Development Board (HDB) plays a central role in the country's residential property market. According to the Singapore Department of Statistics (n.d.), 77.2% of resident households lived in HDB flats in 2025, while the homeownership rate among resident households reached 91.2%. These figures demonstrate the substantial importance of public housing within Singapore's residential landscape. Consequently, understanding HDB resale price movements is relevant not only to individual buyers and sellers but also to property consultants and analysts who require reliable market information to support pricing decisions.

The HDB resale market is characterized by considerable transaction activity and changing price conditions. According to the Housing & Development Board (2026a), annual resale transactions declined from 28,986 in 2024 to 26,169 in 2025, representing a decrease of 9.7%. Meanwhile, annual growth in the HDB Resale Price Index (RPI) moderated from 9.7% in 2024 to 2.9% in 2025, with the index reaching 203.6 in the fourth quarter of 2025. These developments indicate that market activity and price movements can change substantially over time. Therefore, relying exclusively on historical market averages without considering transaction timing may provide an incomplete basis for evaluating individual property prices.

Beyond market-wide fluctuations, substantial resale price differences exist across geographical locations and flat categories. Official HDB statistics for the fourth quarter of 2025 reported a median resale price of SGD 540,000 for four-room flats in Jurong West, compared with SGD 998,000 for the same flat category in Queenstown, representing a difference of approximately 84.8%. Similarly, median prices for four-room flats were SGD 555,000 in Woodlands and SGD 922,000 in Bukit Merah (Housing & Development Board, 2026). These differences highlight the limitations of applying a single market-wide average when evaluating properties with different characteristics. Nevertheless, such comparisons are descriptive and do not establish that location alone explains the observed price differences.

At the transaction level, HDB resale prices may also be associated with property characteristics such as town, flat type, floor area, storey range, flat model, remaining lease, and transaction period. The official resale transaction dataset published through data.gov.sg contains 11

attributes, including actual resale prices and relevant property characteristics (Housing & Development Board, 2021). Importantly, HDB notes that published resale prices should be regarded as indicative because prices negotiated between buyers and sellers depend on multiple factors. Accordingly, a more systematic analytical approach is needed to examine the relationships between property characteristics and transaction prices rather than relying solely on aggregated historical averages.

Against this background, the central research problem addressed by this project is how historical HDB resale transaction data can be systematically analyzed to estimate property-specific resale prices and provide quantitative evidence for pricing decisions. The project, *Predicting HDB Resale Flat Prices to Support Property Pricing Decisions in Singapore*, addresses this problem by integrating MySQL-based data management, KNIME-based data preparation and exploratory analysis, Linear Regression for price estimation, and interactive data visualization. Through the identification of historical pricing patterns, the development of a predictive model, and the evaluation of prediction errors, the project aims to provide a transparent and reproducible analytical framework to support property pricing assessments. Predicted prices are intended to serve as decision-support references rather than substitutes for professional property valuations.

1.2. Project Objectives and Stakeholders

The main objective of this project is to analyze HDB resale price patterns and develop a predictive model to support data-driven property pricing decisions in Singapore.

The project has three specific objectives:

1. Analyse historical resale transactions to understand price differences across towns, flat types, floor areas, and transaction periods.
2. Develop and evaluate a Linear Regression model to predict resale prices using property characteristics available before pricing decisions.
3. Use historical patterns, model predictions, and prediction uncertainty to develop evidence-based pricing recommendations.

The primary decision-makers are property pricing analysts and real estate consultants. Other relevant stakeholders include property buyers and sellers, real estate agencies, and property market analysts.

The proposed system aims to provide data-based reference prices, support property comparisons, and identify cases that may require additional pricing review.

1.3. Decision Questions

To guide the project, three Decision Questions (DQs) were established:

DQ1 – UNDERSTAND: How do HDB resale prices vary across towns, flat types, floor areas and transaction periods?

DQ2 – PREDICT: How accurately can HDB resale flat prices be predicted using property characteristics available before a pricing decision?

DQ3 – DECIDE: How can predicted resale prices and historical transaction patterns support property pricing recommendations across different flat types and towns?

These questions form a connected decision-making process: understanding historical price patterns, evaluating predictive accuracy, and using analytical results to support practical pricing recommendations.

1.4. Project Scope and Approach

The project uses official HDB resale transaction data covering January 2024 to June 2026. A total of 52,916 transactions from 2024–2025 were used for historical analysis and model development, while 12,226 transactions from January–June 2026 were reserved for independent testing.

After data quality assessment and preparation, the model candidate dataset was divided chronologically into 2024 training data and 2025 validation data. Linear Regression was selected to predict the target variable, resale_price.

The system integrates MySQL for data storage and SQL analysis, KNIME for data preparation, feature engineering, and machine learning, and Tableau for interactive visualization. Analytical

outputs and prediction results are written back to MySQL for evaluation and dashboard development.

The project delivers historical market insights, model performance results, empirical prediction ranges, and three evidence-based recommendations. These outputs are intended to support property pricing decisions rather than replace professional valuation or human judgement.

2. DATASET DESCRIPTION

2.1. Data Source and Overview

This project uses HDB resale transaction data published by Singapore's Housing & Development Board (HDB) through the official data.gov.sg portal. The dataset, titled *Resale Flat Prices Based on Registration Date from Jan-2017 Onwards*, provides information about registered HDB resale transactions, including property characteristics, location, lease information, and resale prices.

The project focuses on transactions recorded between January 2024 and June 2026, covering 65,142 transactions over 30 months. Of these, 52,916 transactions from 2024–2025 were selected for historical analysis and model development, while 12,226 transactions from January–June 2026 were reserved for independent model testing.

The dataset is available under the Singapore Open Data Licence, which permits reuse with appropriate source attribution. The project uses a fixed historical extract to maintain consistency in its analysis and evaluation.

2.2. Dataset Attributes

The original dataset contains 11 attributes. Each record represents one registered HDB resale transaction.

Attribute	Description
month	Month of resale transaction
town	Town where the flat is located
flat_type	Type of HDB flat
block	Residential block number

street_name	Street location
storey_range	Floor-level range
floor_area_sqm	Floor area in square metres
flat_model	HDB flat model
lease_commence_date	Year the lease commenced
remaining_lease	Remaining lease duration
resale_price	Recorded resale price in SGD

Table 1. Original HDB Resale Dataset Attributes

The target variable is resale_price, while relevant property characteristics, including town, flat type, floor area, storey range, flat model, remaining lease, and transaction time, were considered as potential predictors.

An additional transaction_id was generated in MySQL as a primary key for record tracking. It was not part of the original dataset and was not used as a predictive feature.

2.3. Data Scope and Experimental Split

The project followed a chronological data-splitting approach to evaluate model performance over time.

Dataset	Records	Purpose
Original 2024–2025 data	52,916	Historical analysis and data preparation
Model Candidate – 2024	27,349	Model training
Model Candidate – 2025	24,286	Internal validation
Independent Test – Jan–Jun 2026	12,226	Final out-of-time model evaluation

Table 2. Data Allocation for Analysis and Modelling

Following the Week 2 data quality assessment, 51,635 transactions from 2024–2025 were classified as model candidates, while 1,281 transactions were flagged for further review. The original records were preserved.

The 2026 dataset was kept separate from model development to assess how well the trained model performed on later transactions. Detailed data quality findings and preparation decisions are presented in Section 3.

3. DATA QUALITY ASSESSMENT

3.1. Data Quality Assessment Methods

Before developing the prediction model, the team assessed the quality of 52,916 HDB resale transactions from 2024 to 2025. The assessment was conducted using SQL in MySQL and data preparation workflows in KNIME.

Five data quality checks were performed: missing values, duplicate records, invalid values, inconsistent values, and outliers. These checks aimed to identify potential data issues and support appropriate preparation decisions without unnecessarily modifying the original transactions.

3.2. Data Quality Assessment Results

The assessment results are summarized in Table 3.

Data Quality Check	Result
1. Missing Values	0 missing values
2. Duplicate Records	27 groups; 28 extra identical rows
3. Invalid Values	0 invalid values under the defined rules
4. Inconsistent Values	1 lease-related case requiring verification
5. Outliers	132 floor area and 1,124 resale price outlier flags

Table 3. Data Quality Assessment Results

The dataset contained no missing values and no invalid values under the applied validation rules. The duplicate assessment identified 27 groups of identical transactions, containing 55 records in total, including 28 extra identical rows.

One transaction showed a potential inconsistency between the lease commencement year and the reported remaining lease. This case was flagged for verification rather than automatically corrected.

Outliers were identified using the Interquartile Range (IQR) method with a 1.5 × IQR threshold. The analysis flagged 132 transactions based on floor area and 1,124 based on resale price. However, unusual values do not necessarily indicate incorrect transactions.

Overall, 1,281 unique transactions were flagged for further review. Some quality flags overlapped, so the individual flag counts should not be added together.

3.3. Data Preparation and Before–After Comparison

Instead of automatically deleting duplicates or outliers, the team adopted a flag-and-review approach. KNIME was used to create quality indicators and review decisions while preserving the original dataset.

Indicator	Before	After
Original transactions retained	52,916	52,916
Records corrected in raw data	—	0
Records deleted from raw data	—	0
Model Candidate Dataset	Not separated	51,635 (97.58%)
Review Dataset	Not separated	1,281 (2.42%)

Table 4. Before and After Data Preparation

After preparation, 51,635 transactions were classified as model candidates, while 1,281 were separated for further review. This separation did not remove records from the original MySQL table.

The model candidate dataset was prepared for subsequent Linear Regression modelling. However, excluding flagged transactions from the modelling dataset may affect its representativeness, particularly for unusual or high-priced properties.

To support transparency and reproducibility, the team documented the cleaning decisions and wrote a seven-category data quality summary back to MySQL in the table hdb_resale_quality_summary_w2.

This approach preserved the original data, provided a clear record of preparation decisions, and established a documented dataset for the next modelling stage.

4. SYSTEM ARCHITECTURE AND DATA PIPELINE

4.1. System Architecture

The project developed an integrated data-driven decision-support system using MySQL, KNIME Analytics Platform, and Tableau. These tools were connected to support data storage, quality assessment, predictive modelling, visualization, and property pricing decisions.

The system follows six main stages: data collection, MySQL storage, KNIME data preparation, Linear Regression modelling, MySQL results integration, and interactive dashboard development. This architecture connects historical transaction analysis with model evaluation and evidence-based decision recommendations.

4.2. Data Processing and Integration

Stage 1 – Data Storage in MySQL

HDB resale transaction data from 2024 to 2025 was imported into the MySQL database dddm_hdb, under the table hdb_resale_raw. The table contains 52,916 transactions, with an additional transaction_id serving as the primary key. SQL queries were used to explore the data, calculate key indicators, and assess data quality.

Stage 2 – Data Preparation in KNIME

KNIME was connected to MySQL to retrieve transaction data for quality assessment, transformation, and exploratory analysis. Five data quality checks were performed, and quality flags were created to support review decisions. The preparation process identified 51,635 model candidate records and 1,281 records requiring further review. A seven-category quality summary was written back to the MySQL table hdb_resale_quality_summary_w2.

Stage 3 – Machine Learning and Evaluation

The prepared data was used to develop a Linear Regression model in KNIME. The model was trained using 2024 transactions and validated using 2025 transactions. Data from January to June 2026 was reserved for independent testing.

Model performance was evaluated using MAE, RMSE, and R^2. Predictions and error measurements were generated to assess accuracy and identify differences in model reliability across property segments.

Stage 4 – Results Integration and Visualization

Prediction results were written back to MySQL, including the 2025 validation predictions in hdb_resale_predictions_w3_2025 and the 2026 independent test predictions in hdb_resale_predictions_w4_2026. Additional comparison and prediction-range outputs were also prepared for analysis.

Tableau was used to visualize historical resale patterns, model performance, prediction bias, and empirical prediction ranges. Interactive dashboards enabled users to explore results across towns and flat types, providing evidence for the three decision recommendations.

4.3. Reproducibility and System Documentation

To support reproducibility, the project maintained SQL scripts, KNIME workflow files, the Tableau workbook, and a README document explaining the system components.

The workflows and analytical outputs were retained for final demonstration and review. Reproducing the system on another computer requires the relevant datasets, database setup, file paths, and MySQL connections to be configured correctly.

5. EXPLORATORY DATA ANALYSIS

5.1. Overview of Historical Resale Transactions

Exploratory Data Analysis (EDA) was conducted on 52,916 HDB resale transactions from 2024 to 2025 to understand historical pricing patterns. The analysis focused on transaction volume, average resale prices, towns, flat types, floor areas, and monthly price trends. SQL, KNIME, and Tableau were used to calculate key indicators and visualize the results.

5.2. Transaction Volume and Price Trends

The dataset contained 27,832 transactions in 2024 and 25,084 in 2025, representing a 9.87% decrease in transaction volume. However, the average resale price increased by 6.52% over the same period.

The monthly price trend also showed a generally upward movement, although prices fluctuated between months. These findings suggest that transaction activity and average resale prices did not follow the same direction during the study period.

5.3. Price Differences by Town and Flat Type

Historical resale prices varied considerably across towns. BUKIT TIMAH recorded the highest average resale price at approximately SGD 920,750, while JURONG EAST had the lowest average of approximately SGD 542,167.

Differences were also observed across flat types. MULTI-GENERATION flats recorded the highest average price of approximately SGD 1,089,289. Among common flat types, 5 ROOM flats averaged SGD 753,757, compared with SGD 648,623 for 4 ROOM flats. The 4 ROOM category accounted for 22,831 transactions, representing approximately 43.15% of the dataset.

These results indicate that location and flat type are relevant factors when comparing historical resale prices. However, average differences do not establish the independent effect of each characteristic.

5.4. Floor Area and Key Insights

The floor area analysis showed that larger flats generally had higher average resale prices. Nevertheless, some large floor area categories contained unusually high values, which should be interpreted carefully because transaction counts may differ across categories.

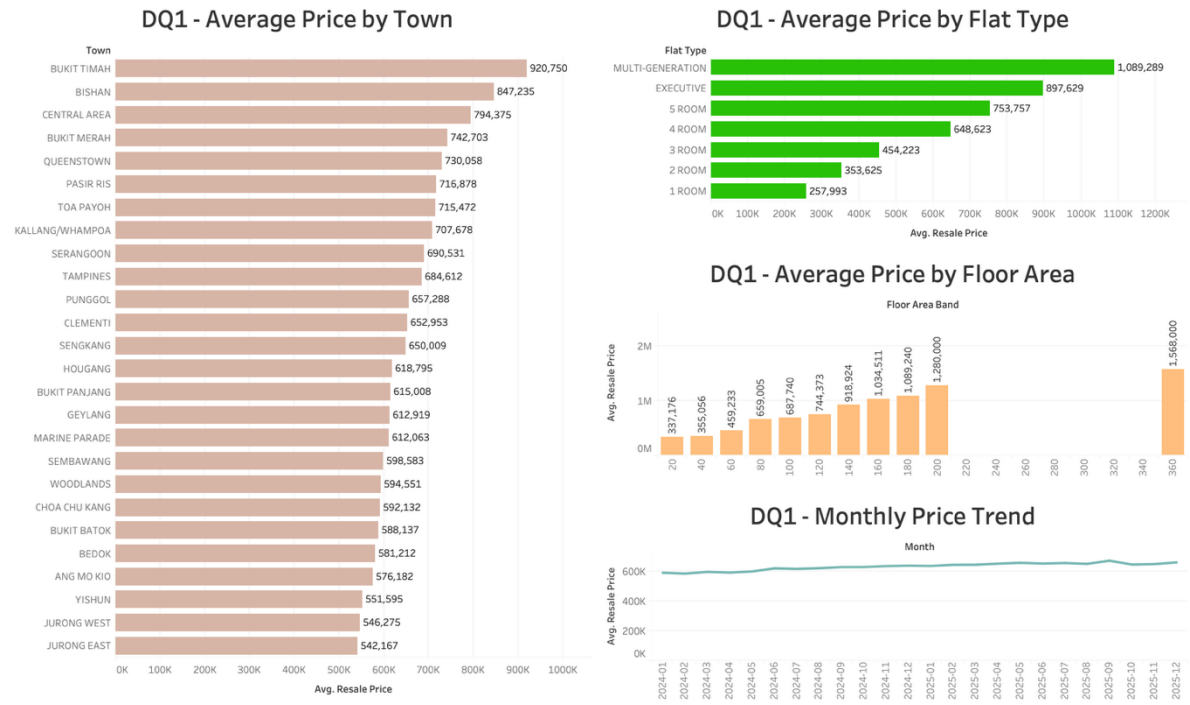


Figure 1: Historical HDB resale price patterns by town, flat type, floor area, and transaction month (2024–2025).

Overall, the EDA findings highlight substantial differences in resale prices across property characteristics and transaction periods. These patterns support the selection of relevant predictors for Linear Regression and demonstrate why property pricing decisions should consider individual flat characteristics rather than relying only on overall market averages.

The findings provide the descriptive foundation for DQ1 – UNDERSTAND, while the predictive relationships and model accuracy are examined in Section 6.

6. MACHINE LEARNING AND MODEL EVALUATION

6.1. Model Selection and Feature Preparation

Linear Regression was selected to predict HDB resale prices because the target variable, resale_price, is continuous. The model estimates resale prices based on available property characteristics and provides a quantitative basis for supporting pricing decisions.

Following the Week 2 data preparation process, 51,635 transactions were classified as model candidates. Relevant predictors considered during feature preparation included town, flat type, flat model, floor area, storey range, remaining lease duration, and transaction time.

Categorical attributes were transformed into numerical variables using One-to-Many encoding in KNIME. The variable price_per_sqm was excluded because it is calculated using the actual resale price and could introduce target leakage. Similarly, transaction_id was retained for record tracking but excluded from model training.

6.2. Experimental Design and Data Splitting

The project adopted a chronological approach to model development and evaluation. This allowed the model to learn from earlier transactions and be evaluated using later observations.

Dataset	Period	Transactions	Purpose
Training	2024	27,349	Model training
Validation	2025	24,286	Model evaluation
Independent Test	Jan–Jun 2026	12,226	Out-of-time evaluation

Table 5. Training, Validation and Independent Test Datasets

The Linear Regression model was trained using the 2024 model candidate records and evaluated on the 2025 validation data. The One-to-Many transformation learned from the training data was applied to the validation dataset to maintain consistent feature encoding.

The January–June 2026 data was kept separate from the training and validation stages. This separation helped reduce the risk of information leakage from future transactions into model development.

6.3. Model Performance and Baseline Comparison

Model performance was evaluated using three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

The Linear Regression model was also compared with a Town-and-Flat-Type Median Baseline to determine whether it provided more accurate predictions than a simpler historical pricing method.

Evaluation Metric	Validation 2025	Independent Test 2026
Transactions	24,286	12,226
MAE (SGD)	54,923.05	59,316.49

RMSE (SGD)	72,639.19	88,412.19
R^2	0.830	—
Underprediction Rate	74.49%	63.88%

Table 6. Linear Regression Evaluation Results

On the 2025 validation dataset, Linear Regression achieved an MAE of SGD 54,923.05 and an RMSE of SGD 72,639.19, with an R^2 of 0.830.

The Median Baseline recorded an MAE of SGD 76,260.78 and an RMSE of SGD 110,101.59. Therefore, Linear Regression reduced MAE by approximately 28% compared with the selected baseline.

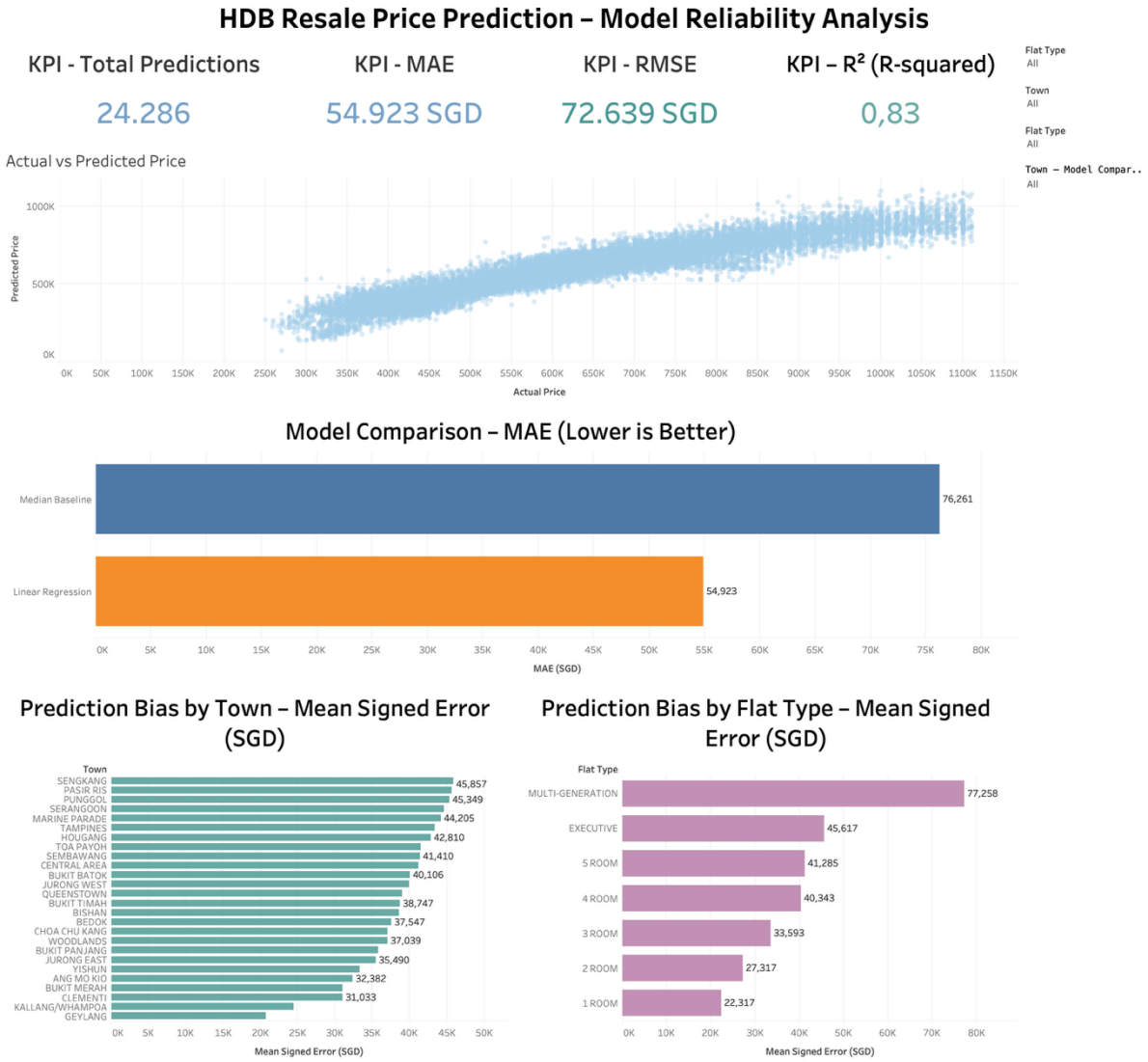


Figure 2: Linear Regression performance and comparison with the Median Baseline on the 2025 validation dataset.

6.4. Independent Test and Model Reliability

The independent test on 12,226 transactions from January to June 2026 produced an MAE of SGD 59,316.49 and an RMSE of SGD 88,412.19.

Compared with the 2025 validation results, MAE increased by approximately 8%, indicating higher average prediction errors on the later dataset. RMSE also increased, suggesting that larger prediction errors had a greater influence on overall performance.

Prediction bias was another important finding. In 2025, 74.49% of predictions were below actual resale prices. This proportion decreased to 63.88% in the 2026 independent test, but systematic underprediction remained a concern.

To communicate prediction uncertainty, the project also calculated an empirical error margin of ±SGD 117,547.79 using the 90th percentile of absolute errors from the 2025 validation data. When applied to the 2026 test results, the estimated prediction ranges achieved 88.34% coverage, slightly below the 90% reference target.

6.5. Interpretation and Limitations

The evaluation results indicate that Linear Regression provided more accurate resale price estimates than the selected Median Baseline on the 2025 validation dataset. However, its higher errors in 2026 demonstrate that predictive performance may vary across different periods.

The model also showed differences in prediction reliability across towns and flat types. Some property segments experienced larger errors, which may affect the usefulness of individual price estimates.

Furthermore, excluding transactions flagged during data quality assessment may limit the model's ability to represent unusual or high-priced properties.

Therefore, model predictions should be interpreted as reference estimates rather than exact property valuations. These findings provide the predictive evidence for DQ2 – PREDICT and support the risk-aware pricing recommendations discussed in Section 8.

7. DASHBOARD AND INSIGHTS

7.1. Interactive Dashboard Design

The project developed interactive Tableau dashboards to present historical HDB resale price patterns, Linear Regression performance, and prediction reliability. The dashboards use analytical and prediction results stored in MySQL, connecting the data processing pipeline with practical decision-making.

The visualizations include key performance indicators (KPIs), bar charts, scatter plots, and comparisons across property segments. Interactive filters allow users to explore results by town

and flat type.

The dashboards were designed to address the three decision questions: understanding historical price patterns (DQ1), evaluating predictive performance (DQ2), and identifying information relevant to pricing decisions (DQ3).

7.2. Historical Price and Model Performance Insights

The historical analysis dashboard showed substantial differences in average resale prices across towns, flat types, and floor areas. These findings highlight the importance of comparing properties with similar characteristics rather than relying only on general market averages.

The Model Reliability dashboard presented actual versus predicted prices, evaluation metrics, and a comparison between Linear Regression and the Median Baseline. The results showed that Linear Regression provided lower prediction errors than the selected baseline on the 2025 validation dataset.

7.3. Prediction Reliability and Uncertainty in 2026

The independent test results revealed differences in prediction reliability across property segments. The 2026 Model Reliability dashboard showed that 63.88% of predictions were below actual resale prices.

Prediction bias also varied by town and flat type. For example, CENTRAL AREA recorded a mean signed error of approximately +SGD 173,939, while 5 ROOM flats recorded approximately +SGD 47,191. Positive values indicate that actual resale prices were higher than predicted prices.

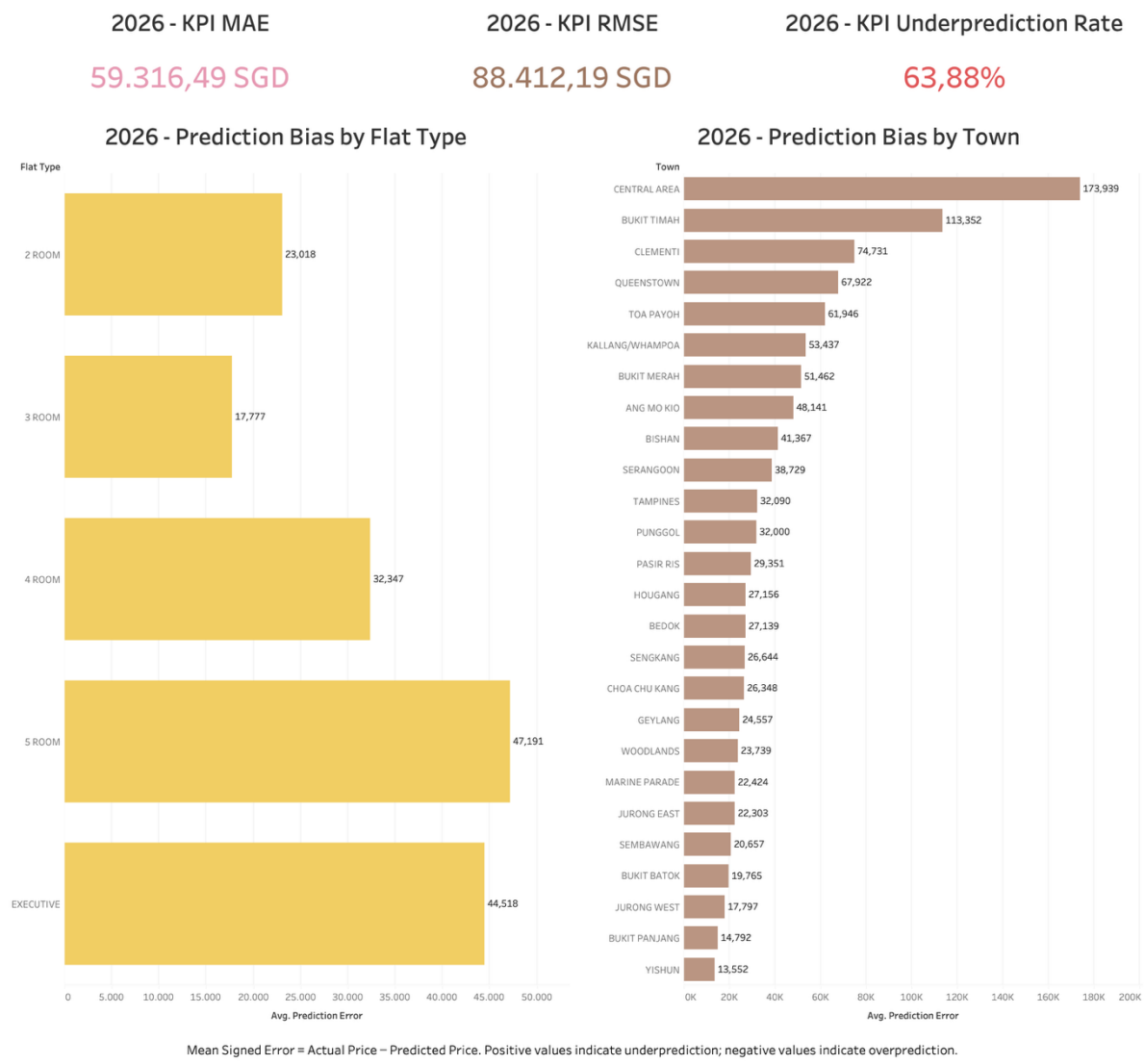


Figure 3: Prediction errors and bias across towns and flat types in the January–June 2026 independent test dataset.

The Prediction Interval dashboard further examined pricing uncertainty. Using an empirical error margin of ±SGD 117,547.79, the estimated price ranges covered 88.34% of actual transactions in the 2026 test dataset, below the 90% reference target. Coverage also differed across towns, with CENTRAL AREA recording only 57.14%.

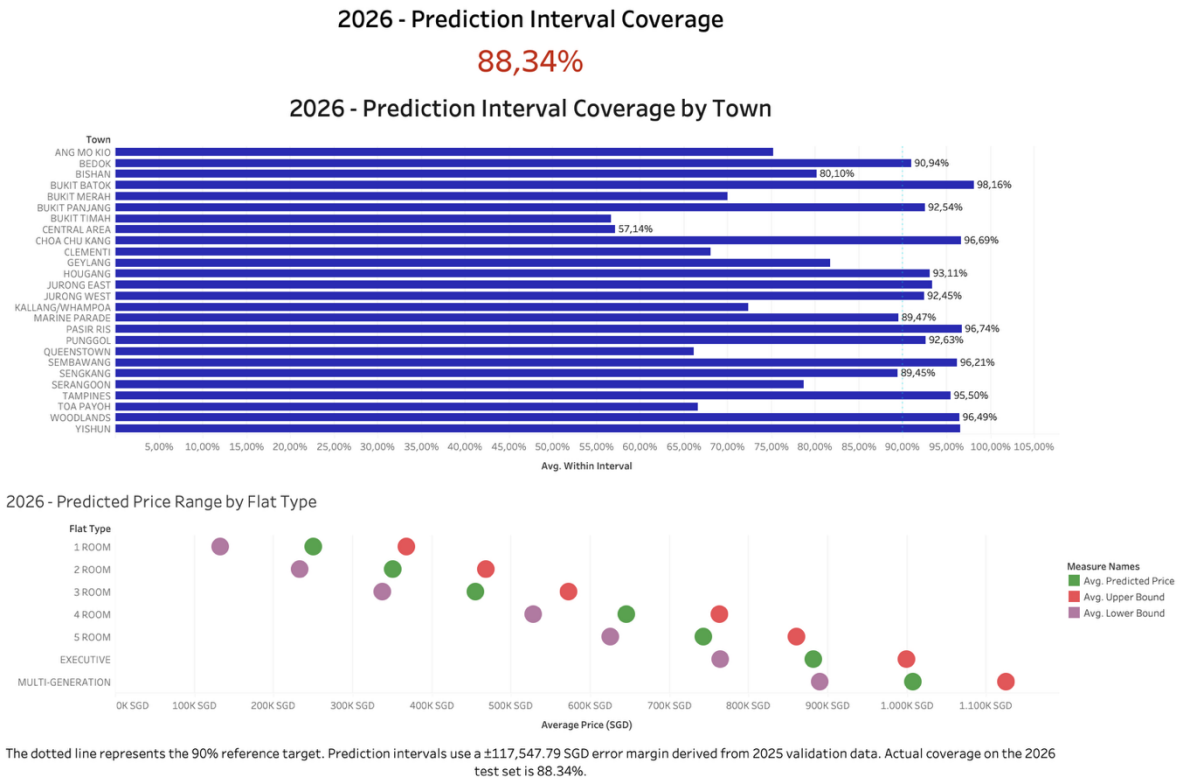


Figure 4: Empirical prediction range coverage by town and estimated price ranges by flat type in 2026.

7.4. Implications for Pricing Decisions

Together, the dashboards show that predicted prices can provide useful reference information, but their reliability differs across property segments.

These findings suggest that pricing analysts should consider historical transactions, prediction uncertainty, and subgroup-level model performance before relying on individual price estimates.

The dashboard insights provide the evidence for the three practical recommendations discussed in Section 8.

8. DECISION RECOMMENDATIONS

This section presents three evidence-based recommendations developed from historical HDB resale transactions, Linear Regression model evaluation, and prediction reliability analysis.

The recommendations aim to support property pricing analysts in making more informed decisions while recognising the uncertainty and limitations of model predictions.

8.1. Recommendation 1: Use Linear Regression and Historical Transactions to Support Pricing Decisions

Recommended Action

Property pricing analysts should use Linear Regression predictions as an initial reference when estimating HDB resale prices. Each predicted price should be compared with historical transactions involving similar towns, flat types, floor areas, and other relevant property characteristics before making a pricing recommendation.

Supporting Evidence

Historical analysis of 52,916 transactions from 2024–2025 showed substantial price differences across property segments. For example, BUKIT TIMAH recorded an average resale price of SGD 920,750.17, compared with SGD 542,167.21 in JURONG EAST.

Model evaluation also demonstrated the value of predictive analysis. On the 2025 validation dataset, Linear Regression achieved an MAE of SGD 54,923.05, compared with SGD 76,260.78 for the Town-and-Flat-Type Median Baseline. This represented approximately a 28% reduction in average absolute prediction error.

Expected Benefit

Combining model predictions with historical transaction evidence can provide a more systematic starting point for pricing decisions. It may help analysts identify proposed prices that require further investigation.

Assumptions and Limitations

This recommendation assumes that the input property information is accurate and that historical price relationships remain relevant. However, the model still produces prediction errors and has shown a tendency to underestimate resale prices. Therefore, predicted prices should support, rather than replace, professional valuation and human judgement.

8.2. Recommendation 2: Use Prediction Ranges to Communicate Pricing Uncertainty

Recommended Action

An estimated price range should be displayed alongside each Linear Regression point prediction. This allows users to consider possible prediction errors instead of treating a single predicted price as an exact valuation.

Supporting Evidence

The project calculated an empirical error margin of ±SGD 117,547.79 using the 90th percentile of absolute prediction errors from the 2025 validation data.

For a predicted price, the estimated range was calculated as follows:

Lower Bound = max(0, Predicted Price − SGD 117,547.79)

Upper Bound = Predicted Price + SGD 117,547.79

When applied to 12,226 transactions in the January–June 2026 independent test dataset, these ranges achieved 88.34% coverage, slightly below the 90% reference target.

For example, a predicted price of SGD 600,000 would produce an estimated range of approximately SGD 482,452 to SGD 717,548.

Expected Benefit

Providing a price range can help analysts communicate uncertainty more clearly, reduce overconfidence in individual model predictions, and support more cautious pricing discussions.

Assumptions and Limitations

This approach assumes that historical prediction errors remain informative for later transactions. However, the fixed margin does not account for different levels of uncertainty across towns or flat types. The observed coverage was also below the reference target.

These ranges are empirical error-based estimates, not statistically guaranteed 90% prediction intervals. Future improvements could consider segment-specific error margins.

8.3. Recommendation 3: Prioritize Manual Review for High-Risk Towns and Flat Types

Recommended Action

Property pricing analysts should prioritise manual review of predictions associated with towns or flat types that have shown relatively high prediction errors or low empirical prediction range coverage.

Manual review should involve comparing the predicted price with recent transactions involving similar properties and checking relevant property characteristics.

Supporting Evidence

The independent test conducted on 12,226 transactions from January to June 2026 showed differences in model reliability across property segments.

The overall 2026 MAE was SGD 59,316.49, and 63.88% of predictions were lower than actual resale prices.

CENTRAL AREA recorded a mean signed error of approximately +SGD 173,939 and an empirical prediction range coverage rate of only 57.14%. Among flat types, 5 ROOM flats showed a mean signed error of approximately +SGD 47,191.

These findings suggest that some property segments may face a greater risk of systematic underprediction or unreliable price estimates.

Expected Benefit

A risk-based review process may help analysts identify less reliable estimates before making pricing decisions. It could also improve the allocation of review effort by focusing attention on property segments where the model has shown greater difficulties.

Assumptions and Limitations

This recommendation assumes that past subgroup performance provides useful information about future prediction risks. However, segment-level results can be affected by differences in

transaction counts and property characteristics. High historical errors do not mean every future prediction within the same segment will be inaccurate.

Manual review also requires additional time and resources, and its effectiveness has not yet been tested in this project.

8.4. Overall Decision Implications

Together, the three recommendations provide a practical approach to data-driven HDB resale pricing decisions.

First, Linear Regression predictions can serve as initial reference prices when combined with historical transaction evidence. Second, empirical prediction ranges can help communicate uncertainty. Third, subgroup-level model reliability can guide the prioritisation of manual reviews.

These recommendations connect DQ1 – UNDERSTAND, DQ2 – PREDICT, and DQ3 – DECIDE by translating historical patterns and predictive results into practical decision support.

However, the proposed system is not intended to produce definitive property valuations. Its performance should be monitored and reassessed as new transaction data becomes available.

9. CONCLUSION AND LIMITATIONS

9.1. Conclusion

This project developed a data-driven decision-support system for analysing and predicting HDB resale flat prices in Singapore. By integrating MySQL, KNIME, Linear Regression, and Tableau, the team established a workflow connecting data preparation, predictive modelling, model evaluation, and decision-making.

The analysis of 52,916 transactions from 2024–2025 identified differences in resale prices across towns, flat types, floor areas, and transaction periods. Following data quality assessment, 51,635 transactions were classified as model candidates, while 1,281 were flagged for further review without modifying the original records.

The Linear Regression model achieved an R^2 of 0.830 and an MAE of SGD 54,923.05 on the 2025 validation dataset, outperforming the selected Median Baseline. On the independent January–June 2026 test dataset, the model recorded an MAE of SGD 59,316.49, while the empirical prediction ranges achieved 88.34% coverage.

These results supported three practical recommendations: using predicted prices alongside historical transactions, communicating uncertainty through estimated price ranges, and prioritising manual reviews for higher-risk property segments. Together, these recommendations demonstrate how historical and predictive analysis can support more informed property pricing decisions.

9.2. Limitations and Future Improvements

Despite these findings, the project has several limitations. First, the dataset does not include all factors that may influence resale prices, such as interior condition, renovation quality, and proximity to transport facilities.

Second, excluding transactions flagged during data quality assessment may reduce the model's ability to represent unusual or high-priced properties. Model accuracy also varied across towns and flat types, and the 2026 independent test covered only six months.

Third, the fixed empirical prediction margin did not achieve the 90% reference coverage target and may not reflect different levels of uncertainty across property segments.

Future improvements could include incorporating additional property characteristics, reviewing the effect of flagged transactions, updating the model with newer data, and developing segment-specific prediction ranges. Further testing would also be needed to evaluate the practical effectiveness of the proposed pricing recommendations.

Overall, the system should be used as a decision-support tool rather than a replacement for professional property valuation. {Citation}

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